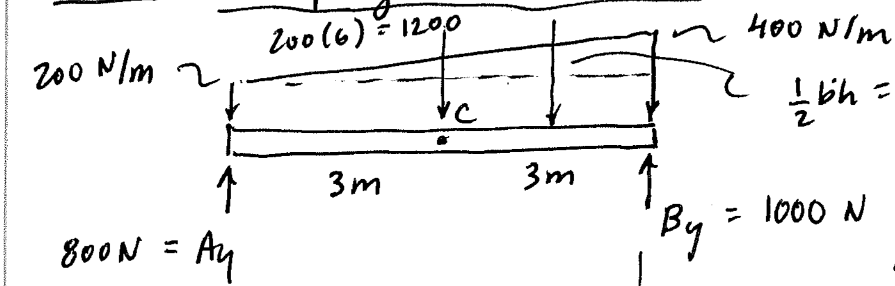
7-16 Trapezoidal Load

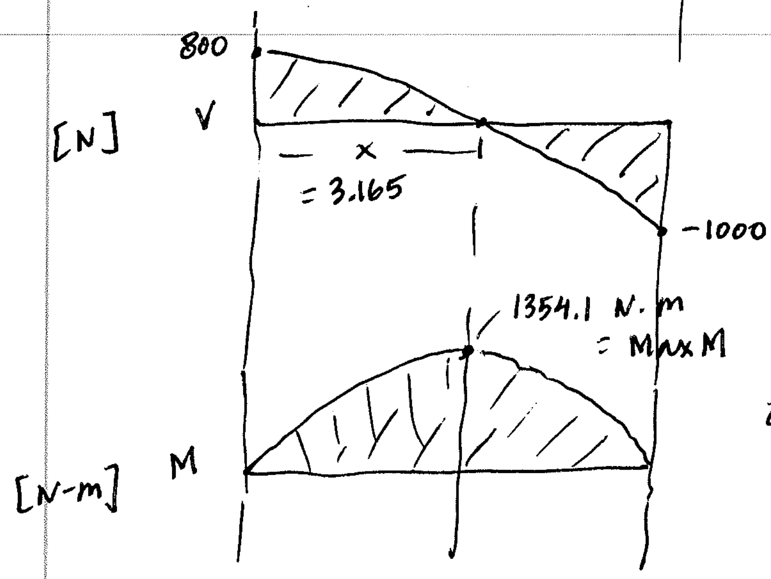


$\frac{1}{2}bh = \frac{1}{2}(6)(400-200)$
$= 600$ N

$A_y = \frac{1}{2}1200 + \frac{1}{3}(600)$

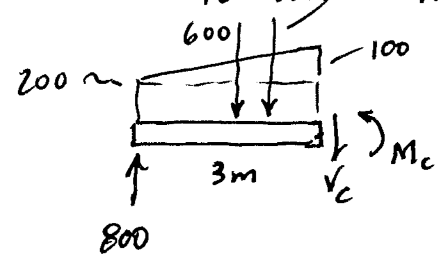
$A_y = 600 + 200$
$= 800$ N

$B_y = 1000$ N



$\frac{1}{2}(3)(100) = 150$

To work the problem:



$+\uparrow \quad 800 - 600 - 150 - V_c = 0$

$\boxed{V_c = +50 \text{ N}}$

$+\circlearrowleft \Sigma M_{cut} = 0; \quad M_c - 800(3) + 600(1.5) + 150(1) = 0;$

$M_c = 2400 - 900 - 150 = 0$

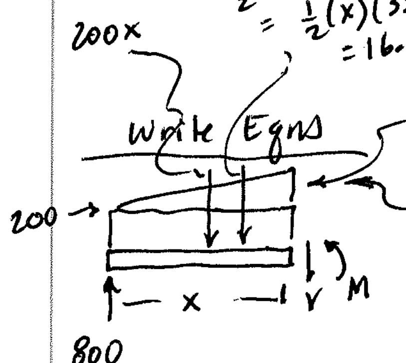
$\boxed{M_c = 1350 \text{ N-m}}$

Write Eqns

$\frac{1}{2}bh = \frac{1}{2}(x)(33.33x) = 16.67x^2$

$\frac{W}{L} = \frac{(400-200)}{6} = 33.33$

$h = 33.33x$



Use Excel Goal Seek

$+\uparrow \quad 800 - 200x - 16.67x^2 - V = 0$

$\boxed{V = 800 - 200x - 16.67x^2}$

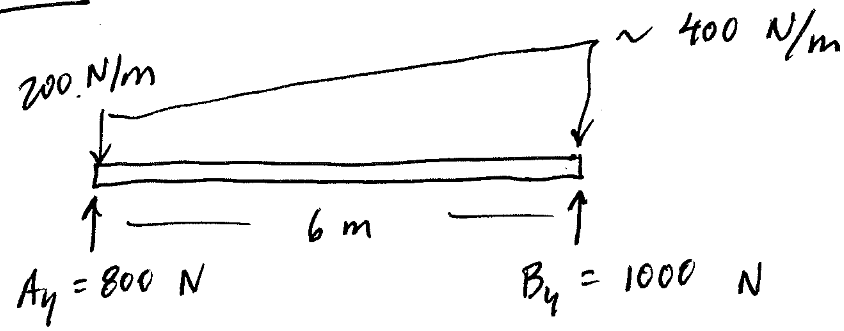
Use Excel: V=0 at $\boxed{x = 3.165 \text{ m}}$

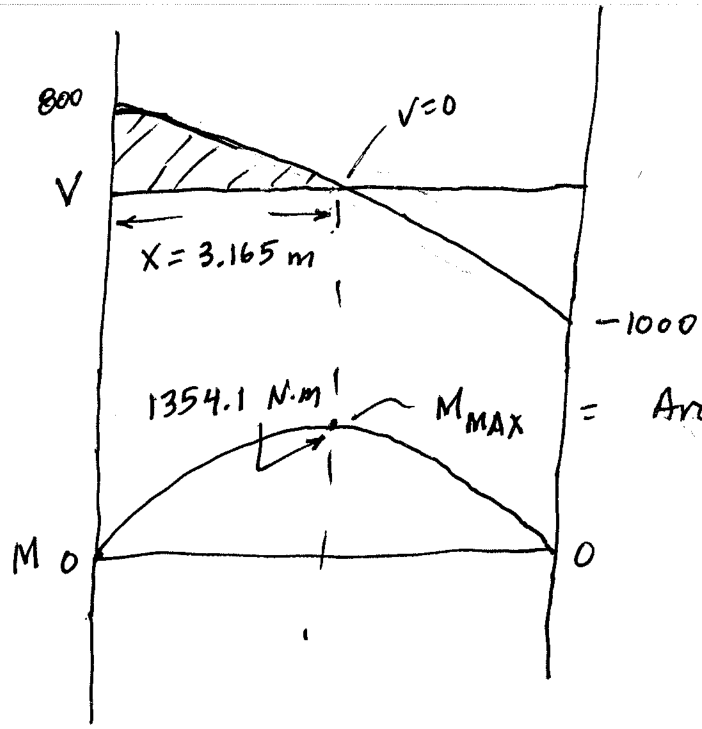
$+\circlearrowleft \Sigma M_{cut} = 0; \quad M - 800(x) + 200x\left(\frac{x}{2}\right) + 16.67x^2\left(\frac{x}{3}\right) = 0$

$\boxed{M = 800x - 100x^2 - 5.56x^3}$

$\boxed{M_{MAX} @ 3.165 = 1354.1 \text{ N·m}}$

7-16





= Area under V curve from
x = 0 to x = 3.165 m
Can integrate V with
TI-89 (see below)

$\underline{0 \le x \le 6}$

$V = 800 - 200x - 16.67x^2$ } Find root: $V = 0$ at $x = 3.165$ m

$\underline{\text{TI-89}}$: zeros(800 - 200 * x - 16.67 * x^2, x)

= -15.163, $\boxed{3.165 \text{ m}}$

Change in M (ΔM) from $x = 0$ to $x = 3.165$
= Area under V curve:

$\underline{\text{TI-89}}$: $\int$(800 - 200 * x - 16.67 * x^2, x, 0, 3.165)

$\boxed{\Delta M = 1354.1}$ N.m